********** MEMS 2026 – SOLUTIONS **********

Solution 1

a) The main transduction mechanisms are:

1) Electrostatic (or capacitive) – based on variation of capacitance between fixed and moving elements.
2) Electromagnetic – based either on inductance variation of a gapped inductor or on EM induction in a coil moving in a static magnetic field generated by an external magnet.
3) Piezoresistive – based on the variation of resistivity with applied mechanical stress.
4) Piezoelectric – based on the direct piezoelectric effect, where applied mechanical stress generates dielectric polarization and surface charge.

Electrostatic and piezoresistive are the most widely used. The reason is that they can be implemented in a silicon mechanical layer without the need for any additional materials. Magnetic and piezoelectric transduction both require the introduction of materials that are difficult to process. Also, it is difficult to make efficient magnetic circuits using MEMS processes. [6]

b) The sidewalls of an anisotropically etched cavity will be{111} planes, while the wafer surface is the (100) plane. The angle between any two planes is $\theta = \cos^{-1}(\mathbf{n1}\cdot\mathbf{n2})$ where **n1** and **n2** are the unit normals. For the planes in question we have **n1** = $(\pm1,\pm1,\pm1)/\sqrt{3}$ and **n2** = (1,0,0), so $\theta = \cos^{-1}(\pm1/\sqrt{3}) = 54.7°$ or 125.3°. Only the acute angle will ever arise because the corresponding plane will be exposed first and will terminate the etch process.

Some {111} group planes are normal to the (110) plane, for example $(1\,\bar{1}\,1)$ and $(\bar{1}\,1\,1)$, and it follows that vertical sidewalls can be obtained with a (110) wafer.

The limiting shapes are a rectangle for (100) and a parallelogram for (110). [6]

c) The transverse stiffness of a cantilever is $k = 3EI/l^3$ where I is the second moment of area and l is the cantilever length. The scaling behaviours of E, I and l are L^0, L^4 and L^1 respectively, so k will scale as L^1. It follows that the elastic force kx will scale as L^2. [4]

The surface tension force scales as L^1 and so can dominate over the elastic force at MEMS scales. This can have negative consequences such as stick-down of moving elements during sacrificial release. On the other hand, it means that surface tension can be used for actuation e.g. in self-assembly processes. [2]

The gravitational force scales as L^3 and so becomes weaker than the elastic force at MEMS scales. This is why MEMS suspensions are characterised by long, slender beams and why gravitational forces can usually be neglected when analysing MEMS devices. [2]

d) A given beam will buckle when the compressive stress reaches a critical value $\sigma_C = P_C/(wh)$ where w is the beam width, h is the polysilicon layer thickness and P_C is the Euler

critical load. Using the equation given in the *Information for Candidates*, and putting $I = h^3w/12$ (because the beam is going to buckle out-of-plane) we obtain:

$$\sigma_C = \left(\frac{2\pi h}{L}\right)^2 E$$

With E = 170 GPa and h = 3 μm, the beam lengths given in the question correspond to stress levels of 10, 20, 30, 40 and 50 MPa. The first two beams are buckled, so we know the stress is compressive and lies in the range 20 to 30 MPa. [6]

e) From the equation given in the Information for Candidates, the self-resonance frequency of the lowest order mode is:

$$f_0 = \frac{3.52}{2\pi}\sqrt{\frac{Ewh^3}{12L^3m}} \quad \text{Hz}$$

where w = 15 μm, h = 5 μm, L = 300 μm are the width, depth and length of the cantilever, $m = 5.48 \times 10^{-11}$ kg is the total mass (including the polymer layer), and E = 170 GPa. The mass is calculated as $m = wL(h\rho_{Si} + t\rho_{Po})$, with ρ_{Si} = 2330 kg/m^3, ρ_{Po} = 1050 kg/m^3 and t = 0.5 μm. Substituting these values gives an initial frequency of f_0 = 75.1 kHz.

Since $f_0 \propto m^{-0.5}$ it follows that the frequency shift due to a small change in mass will be:

$$\Delta f_0 \approx -0.5 f_0 \frac{\Delta m}{m}$$

With $\Delta f_0 \approx -0.5$ Hz, this gives Δm = 0.73 pg. [6]

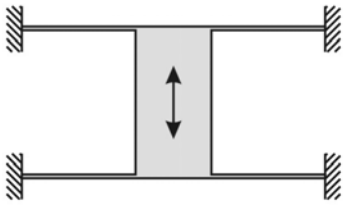
f) Monolithic integration eliminates the need for hybrid packaging which is expensive. It also generally leads to a more compact packaged device. Having the electronics physically close to the MEMS device can also improve performance through, for example, lower susceptibility to noise and interference. [2]

Materials are: A – sacrificial oxide; B – polysilicon; C – CMOS metallisation, e.g. aluminium, D – silicon nitride passivation; CMOS dielectric, e.g. oxide. [4]

Advantages: (1) more freedom over MEMS deposition/annealing conditions in absence of CMOS circuitry; (2) standard CMOS foundry process can be used. [2]

Solution 2

a) Hammock suspension:



Because of the fixed span, the flexures in a hammock suspension become stretched when they are deflected laterally. The resulting axial stress increases the lateral stiffness, making the suspension non-linear.

Non-linearity is undesirable because it leads to jumps and hysteresis in the frequency response, making it difficult to control the oscillation frequency and amplitude near resonance. [6]

b) When the structure is deflected by a y-axis displacement of the proof mass, the spans of the portal frame and the central flexure will both decrease so as to keep their respective arc lengths constant. If the width of the central flexure is chosen so that the reductions in span are equal, then the central flexure will remain stress free. [4]

From the *Information for Candidates* we know that reduction in span is proportional to u^2/L so, if u_f and u_p are the deflections of the flexure and portal frame, we require:

$$\frac{u_f^2}{l_f} = \frac{u_p^2}{l_p}$$

Also, we know that the flexure and the portal frame are carrying the same transverse load, so we can write:

$$k_f u_f = k_p u_p$$

where $k_f = \frac{E w_f^3 h}{l_f^3}$ and $k_p = \frac{2 E w_p^3 h}{l_p^3}$ are the stiffnesses of the flexure and portal frame respectively, h being the mechanical layer thickness. Combining these equations and rearranging to make w_f^3 the subject gives:

$$w_f^3 = 2\left(\frac{l_f}{l_p}\right)^3 \frac{u_p}{u_f} w_p^3 = 2\left(\frac{l_f}{l_p}\right)^{5/2} w_p^3$$ [6]

c) All three flexures exert couples on the cross-bar and these will not sum to zero because of the length difference. The net couple will need to be balanced out by axial forces in the portal frame flexures. [4]

The axial forces will modify the stiffnesses of the portal frame flexures, but since they are equal and opposite, and the variation of flexure stiffness with axial load is close to linear, any effect on the overall stiffness with loading is expected to be negligible. [4]

d) The overall stiffness of the suspension will be:

$$k = \frac{4}{\frac{1}{k_p}+\frac{1}{k_f}},$$

and this needs to satisfy:

$$k = (2\pi f_0)^2 m = (2\pi f_0)^2 \rho A h.$$

Also, using the result in part b), we know that $\frac{w_f}{w_p} = 1.517$.

Combining these relations we can solve for the flexure widths, and this gives $w_p = 4.86$ µm and $w_p = 7.37$ µm. [6]

Solution 3

a) With $\sigma_z = 0$ the first two Hooke's Law equations become:

$$E\varepsilon_x = \sigma_x - \nu\sigma_y$$
$$E\varepsilon_y = \sigma_y - \nu\sigma_x$$

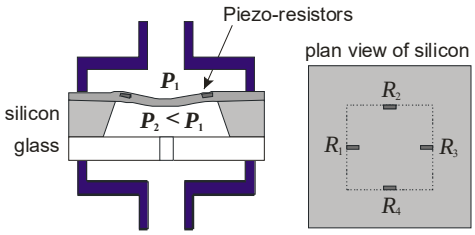
Eliminating σ_y from these equations we get:

$$E\varepsilon_x = \sigma_x - \nu\left(E\varepsilon_y + \upsilon\sigma_x\right) \quad \Rightarrow \quad E\varepsilon_x + \upsilon E\varepsilon_y = \sigma_x(1-\upsilon^2)$$

and the RH equation is trivially rearranged to give the first of the required equations. The second equation is obtained by eliminating σ_x from the starting equations above. [6]

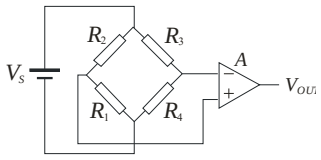
b) Along any edge of the membrane, the strain parallel to the edge will be zero because the adjacent bulk material will force this condition. So, for example, on a y-aligned edge, we will have $\varepsilon_y = 0$. Substituting this into the first equation of the part a) solution, we see that on, on this edge of the membrane, $\sigma_y = \nu\sigma_x$ and, since we are on a y-aligned edge, $\sigma_y \equiv \sigma_p$, $\sigma_x \equiv \sigma_n$, and the quoted result applies. A similar argument leads to the same conclusion for the x-aligned edges. [4]

c) Layout of a typical bulk micromachined pressure sensor:



The membrane is aligned to <110> directions because the standard substrate material is <100>-oriented and the intersections of the slow-etching {111} planes with the surface of such a wafer lie along the <110> directions. [6]

d) The piezoresistors R_1 and R_3 experience longitudinal and transverse stresses $\sigma_L = \sigma_{max}$ and $\sigma_T = \upsilon\sigma_{max}$ respectively, where σ_{max} is the maximum membrane stress. Using the standard form given in the *Information for Candidates*, the fractional resistance changes for these resistors will be $\frac{\Delta R}{R} = (\pi_L + \upsilon\pi_T)\sigma_{max}$. The corresponding result for the piezoresistors R_2 and R_4 is $\frac{\Delta R}{R} = (\upsilon\pi_L + \pi_T)\sigma_{max}$. If we now connect the four resistors to form a bridge as shown:



the output voltage (with $A = 1$) will be:

$$V_{OUT} = \left(\frac{R_1}{R_1+R_2} - \frac{R_4}{R_4+R_3}\right) V_s = \frac{(1-\upsilon)(\pi_L-\pi_T)\sigma_{max}}{2+(1+\upsilon)(\pi_L+\pi_T)\sigma_{max}} V_s \approx \frac{(1-\upsilon)\pi_{44}\sigma_{max}}{2} V_s$$

[6]

e) Substituting for σ_{max} from the Information for candidates into the above equation for V_{out} gives:

$$V_{OUT} \approx \frac{3}{20}(1-\upsilon)\pi_{44}\left(\frac{L}{h}\right)^2 V_s\, p$$

Rearranging to make h the subject of this equation:

$$h = L\sqrt{\frac{3}{20}\frac{(1-\upsilon)\pi_{44}V_s}{S}}$$

where $S = V_s/p$ is the sensitivity. With L = 1 mm, ν = 0.064, $\pi_{44} = 138.1 \times 10^{-11}$ Pa^{-1}, V_s = 5 V, and S = 200 nV/Pa, this gives h = 70 μm. [6]

Rearranging the expression for σ_{max} given in the Information for Candidates:

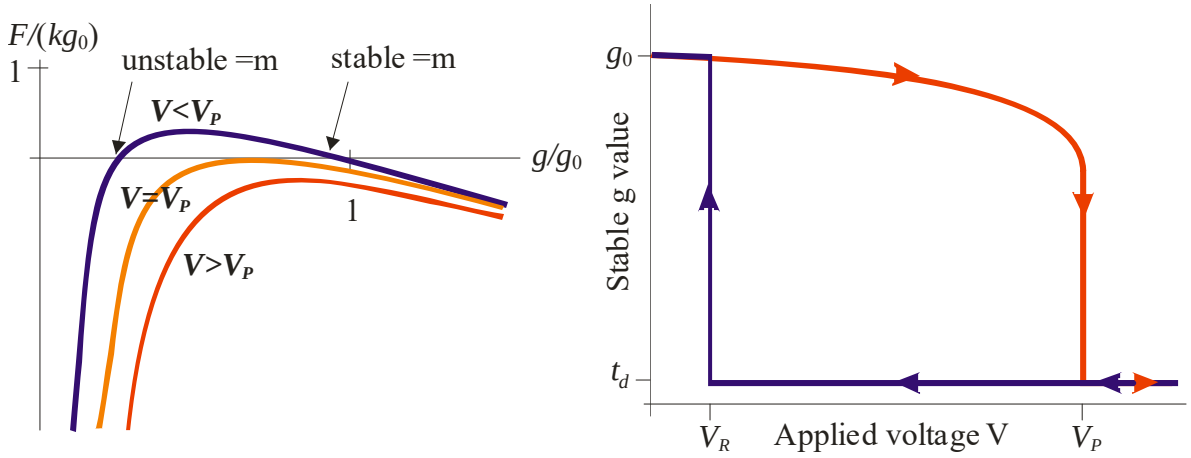
$$p \approx \frac{10}{3}\left(\frac{h}{L}\right)^2 \sigma_{max}$$

With $\sigma_{max} = 200$ MPa this gives $p \approx 3.3$ MPa. [2]

Solution 4

a) The electrostatic force on the moveable plate is $F_e = -A\varepsilon_0 E^2/2$ where $E = V/(g + t_d/\varepsilon_r)$ is the electric field in the gap. The only other force acting on the moveable plate is the spring force $F_k = k(g_0 - g)$, so the total force is:

$$F = F_e + F_k = k(g_0 - g) - \frac{\varepsilon_0 A V^2}{2(g + t_d/\varepsilon_r)^2} \qquad (1)$$



LH graph shows variation of force with gap for different values of V. When $V < V_P$ there are two equilibrium points (i.e. points where $F = 0$). One is stable, and the other is unstable, but the moveable plate naturally settles at the stable one (and never reaches the unstable one) if the applied voltage is increased from zero. For $V > V_P$ there is no equilibrium point, and the total force is always negative so the plate snaps down. Once this has occurred, the voltage must be reduced to the point where the electrostatic force at $g = 0$ falls below the spring force. This results in hysteretic behaviour as shown in RH graph. [8]

b) The conditions at the point of snap-down are $F = 0$ and $\partial F/\partial g = 0$. Differentiating the force equation gives:

$$\frac{\partial F}{\partial g} = -k + \frac{\varepsilon_0 A V^2}{(g + t_d/\varepsilon_r)^3} \qquad (2)$$

So, when $\partial F/\partial g = 0$ we have: $\varepsilon_0 A V^2 = k(g + t_d/\varepsilon_r)^3$ (3)

Substituting (3) into (1), and setting $F = 0$, we find that $g = \frac{2g_0}{3} - \frac{t_d}{3\varepsilon_r}$ at the point of snap-down. Substituting this value of g into (3) gives the quoted result. [8]

c) The release voltage is obtained by setting the total force to zero at g = 0. From (1) this gives:

$$0 = kg_0 - \frac{\varepsilon_0 A V^2}{2(t_d/\varepsilon_r)^2} \quad \Rightarrow \quad V = \sqrt{\frac{2kg_0}{\varepsilon_0 A}}\frac{t_d}{\varepsilon_r} \qquad [4]$$

d) The stiffness of a built-in beam subject to a point load at the centre is $k = 192EI/L^3$, where $I = bd^3/12$ is the second moment of area. (This result is easily derived, either from first principles or from the cantilever result. It may also be quoted from memory.) For the gold bridge we have E = 80 GPa, b = 100 μm, d = 1 μm and L = 500 μm, so we expect the stiffness to be k = 1.02 N/m.

With k = 2 N/m, g_0 = 3 μm, A = 100 x 100 μm^2, the snap-down voltage is obtained as V_P = 9.6 V. This ignores the term t_d/ε_r which makes negligible difference to the answer.

[6]

e) Compared to solid-state RF switches, which are normally based on pin-diodes or FETs, RF MEMS switches generally offer lower on-state insertion loss, better off-state isolation, lower power consumption (assuming electrostatic actuation), and better linearity. [4]

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